TORUS

Paris Gennaro, Jacob Rempel,
Josh Sweet + Will Wingrove



Table of Contents

Design Process	1- 2
B.O.M + Concept drawings	3 - 4
Lab Documentation	5 - 6
Team Discussion	7 - 8

Design Process





When we started this process we instantly liked the idea of a lamp as it allowed us to get to explore light in a way we haven't yet, and to overall create an interesting portfolio piece, functional mood lamp, and exploration of thermoforming techniques simultaneously. The torus shape was excellent for this because the hollow center is both visually interesting and provides insight to the behavior of plastic. Keeping with the simple geometric theme, a triangular prism base creates an interesting form and a flat plane to mount the electronics.

Our final design was ultimately the product of constraints we learned about through the manufacturing process. Our original plan to use tinted acrylic on our vacuum formed housing failed because of the pigments in the acrylic that affect the draw depth of the acrylic which we were unaware of before we started. The result of this was a few catastrophic vacuum forming attempts where the material failed to come close to the desired shape. Our workaround was to substitute tinted acrylic for frosted PETG, which heated up much faster and formed to the shape we needed without much error.

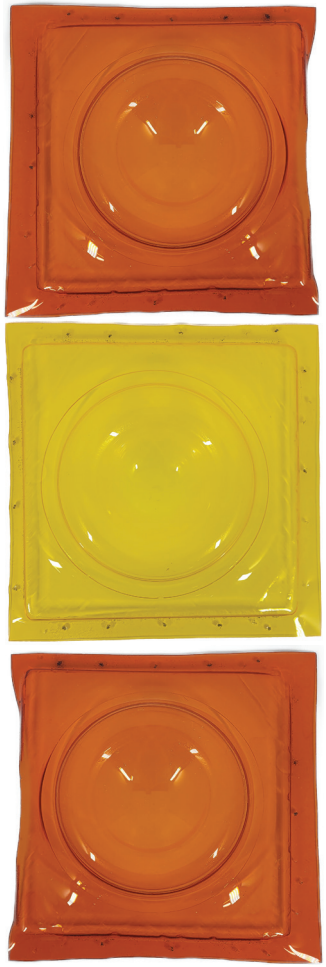
Material → Process → Result

Plywood → Cut and turned on lathe → Tool matches desired shell profile
Plywood → Venting holes drilled → Tool will give desired shape when vacuum forming
Plywood → Cut/Glued → 60 degrees for bending jig

PETG → Vaccum forming → Shells have rough shape
PETG → Cutting → Shells have desired shape
PETG → Sandblasting → Give shells frosted texture
PETG → Gluing → Attach shells

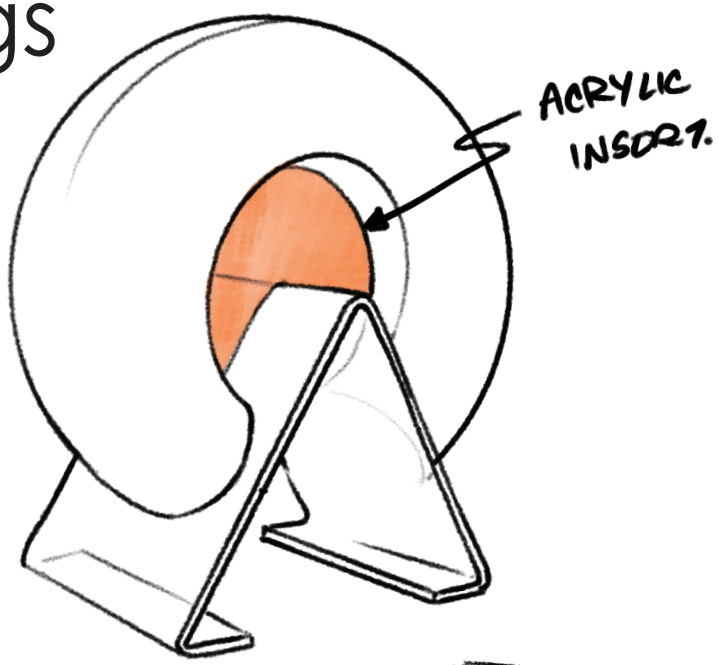
Tinted acrylic → Cutting → Shape and holes for power supply port, switch, and vents, window cut
Tinted acrylic → Heat Forming → Base has desired shape
Tinted acrylic → Gluing → Attach shell to base and window

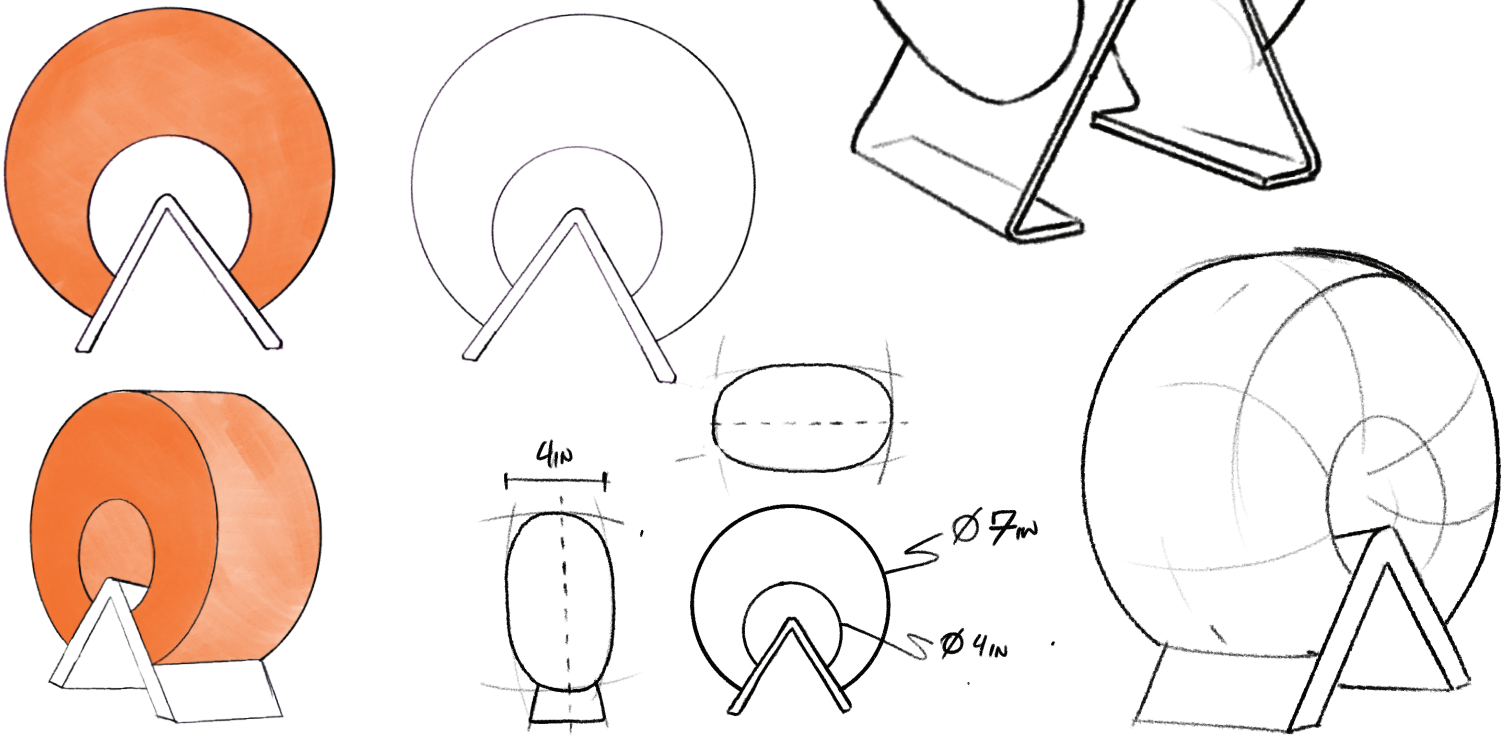
LED light strips → Soldering →

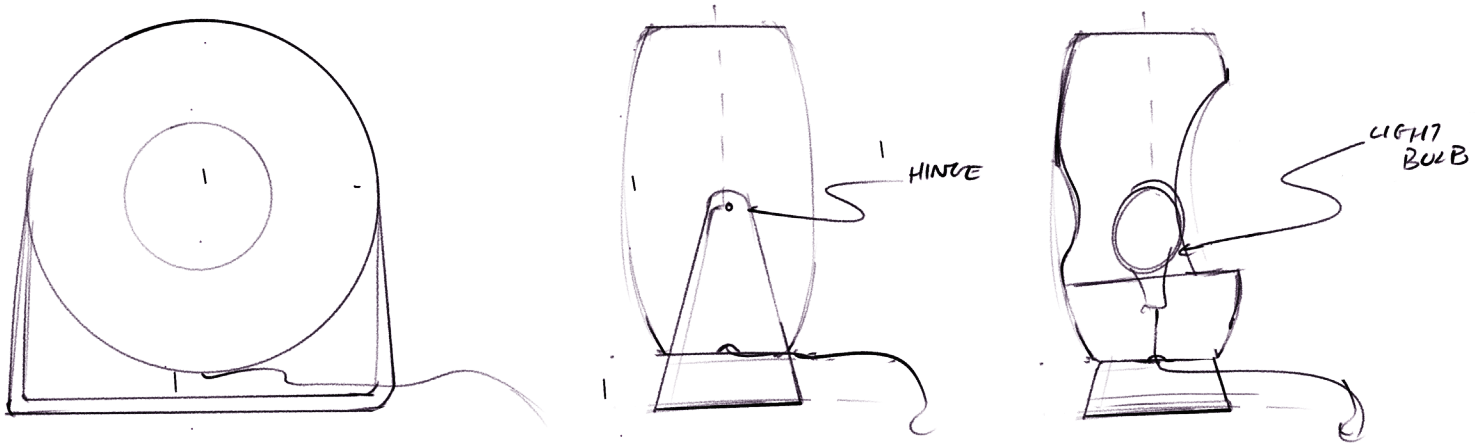


Concept Drawings

The final design was achieved through initial concept sketches that defined the materials and manufacturing processes needed. The development began with a simple prototype made of cardboard and foam that allowed us to achieve the proportions and explored dimensions for stability.







NO	DESCRIPTION	MATERIAL	QTY	COST (SUPPIER)
1	BODY	ACRLYC SHEET	?	? (CANUS)
2	LIGHTS	LED STRIP 8MM NATURAL WHITE 4000K	5M	$22.98 (AMAZON)
3	SWITCHES	ROCKER SWITCH 12V	5	$7.99 (AMAZON)
4	POWER SUPPLY	12V LED POWER SUPPLY	5	$32.99 (AMAZON)

Laboratory Documentation

1. A thick chunk of plywood was cut into a disk and turned on the lathe to match a 3d printed profile template.
2. Holes were drilled into the concave corners of the tool to achieve the desired shape when vacuum forming.
3. With some difficulty, the plastic shells were vacuum formed to the mold.
4. With even more difficulty, the shells were removed from the mold and the desired part was cut free using a dremel.

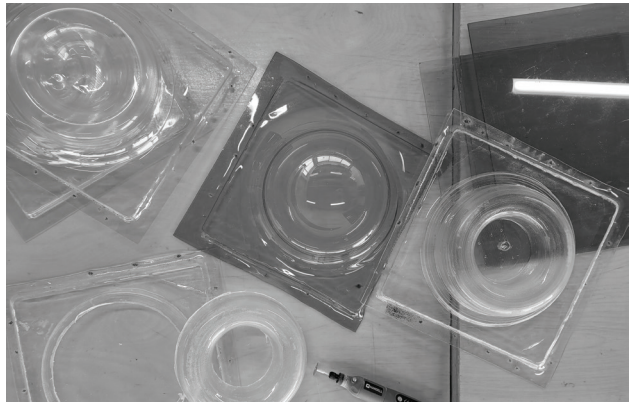




5. The shells were sandblasted to achieve a frosted texture so light would diffuse nicely.
6. The shells were sanded to ensure mating surfaces were flush and glued together with superglue.

7. A bending jig with a 60(degrees symbol) angle was fabricated out of plywood.
8. Tinted acrylic was cut, heated, and bent around the jig to create the triangular base.
9. A dremel was used to cut holes for the power supply port, switch, and vents.





10. The power supply port, switch, and LED light strips were soldered together and threaded through the shell.
11. After some tests, the shell was super glued into position and excess glue was removed.
12. A tinted acrylic window was cut to fit the center to reduce the shadow from the LED strip backside on the inner ring.

Disscusion

Paris Gennaro

Overall our group worked really well together, and we have collaborated in the past with great success so figured it was a good project to bring it back with a more physical objective. We all have very different skills when it comes to hands-on capabilities and this project let us showcase and work with each other's strengths. Most things went really smoothly in terms of group work, though we did fall pretty behind for a while as we were all on and off sick, this forced us to be picking up all the slack we could when we could to make sure everything got done in time. We did end up sacrificing the final finish of the project which definitely could have been a lot cleaner if we had more time. I learned a lot overall since I've never worked with plastics of any sort before this, and it was also a chance for us to learn how to use the lathe which was really interesting.

Jacob Rempel

One corporate adjective describes this crew best. SYNERGY. This project was the second undertaking for our dream team and we obliterated each and every obstacle in our way. We operated as a well oiled machine that consistently turned plague affliction, time constraints, disagreements, and unforeseen problems into learning experiences and exceptional results. Through our grit and determination, we produced a gorgeous and functional lamp prototype. Division of labour was seamlessly and dynamically adjusted to accommodate unforeseen circumstances and morale was kept high even during our lowest moments (the tinted acrylic disaster of 2025). Overall, I would happily work with this group again.

Josh Sweet

Working with this group with very knowledgeable and skilled designers allowed us to work together smoothly and efficiently to develop a clean final prototype. Our diverse skills were divided up and contributed in important areas of the project respectively. The growth of our knowledge throughout the project allowed us to overcome obstacles and unexpected problems, and in the end turn out a spectacular functioning lamp. This project was a first for me that provided valuable experiences in working with thermoplastics and exploring the capabilities of thermoforming. I gained valuable skills in thoughtful use of jigs and tools to ensure accurate forming of the design and easier post-operation processes. Working with lighting in a project was new for me, so this gave me new experience in an area that I have not worked in before.

Will Wingrove

Overall, I think we worked really well as a group. We've collaborated in the past with good results, so it made sense to work together again. This time was no different, everything went smoothly and we didn't run into any major group problems. That said, we did face some challenges. A couple of us were sick for a handful of classes which put us behind schedule. Because of that, we had to adjust and work efficiently to make up for lost time. Another issue we ran into was with our material choice. Tinted acrylic does not like to vacuum form like transparent acrylic or just different types of plastic. Despite that, we managed to get 2 pieces of PETG sheet and used that to vacuum form our parts. Because of that, we had to adjust and work efficiently to make up for lost time. We managed to get back on track and in the end everything worked out.